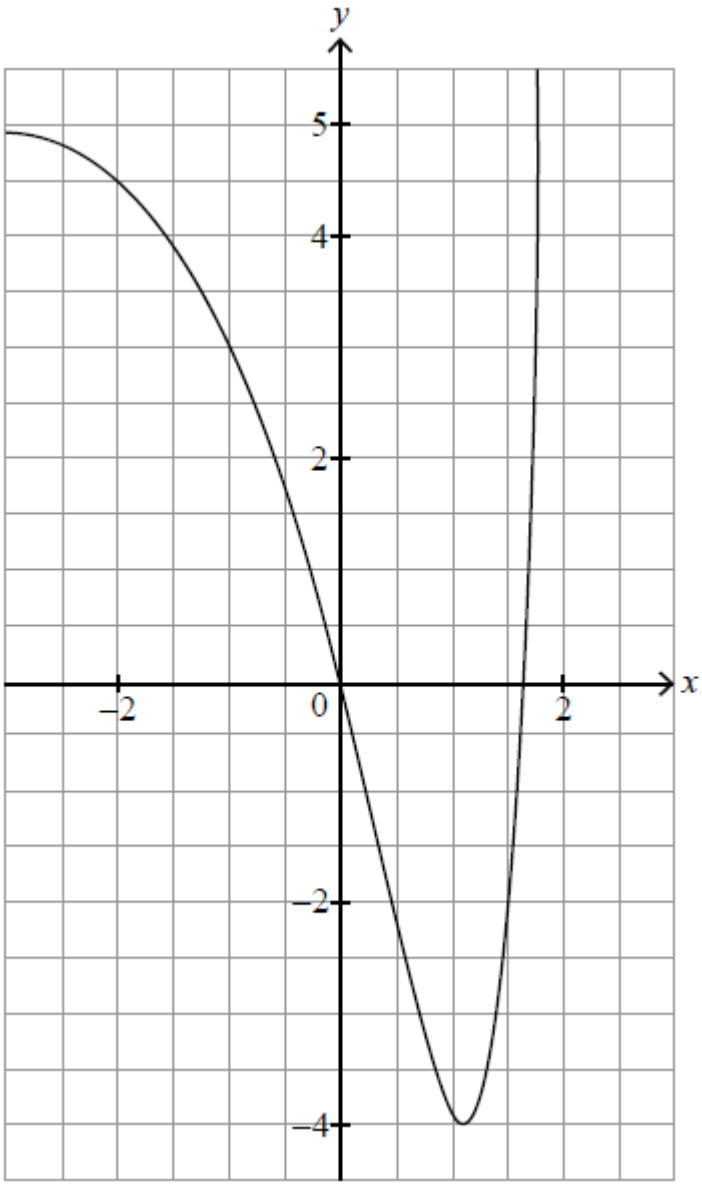
FUNCTIONS AAHL MARCH 26 [91 marks]

1. [Maximum mark: 8]

The function f is defined by $f(x) = e^{2x} - 6e^{x} + 5,\ x \in \mathbb{R},\ x \leqslant a$. The graph of $y = f(x)$ is shown in the following diagram.



(a) Find the largest value of a such that f has an inverse function. [3]

Markscheme

attempt to differentiate and set equal to zero *M1*

$f'(x) = 2e^{2x} - 6e^{x} = 2e^{x}(e^{x} - 3) = 0$ *A1*

minimum at $x = \ln 3$

$a = \ln 3$ *A1*

[3 marks]

(b) For this value of a, find an expression for $f^{-1}(x)$, stating its domain. [5]

Markscheme

Note: Interchanging x and y can be done at any stage.

$y = (e^{x} - 3)^{2} - 4$ *(M1)*

$e^{x} - 3 = \pm\sqrt{y+4}$ *A1*

as $x \leqslant \ln 3, x = \ln\left(3 - \sqrt{y+4}\right)$ *R1*

so $f^{-1}(x) = \ln\left(3 - \sqrt{x+4}\right)$ *A1*

domain of f^{-1} is $x \in \mathbb{R}, -4 \leqslant x < 5$ *A1*

[5 marks]

2. [Maximum mark: 5]

Consider the graphs of $y = \frac{x^2}{x-3}$ and $y = m(x+3), m \in \mathbb{R}$.

Find the set of values for m such that the two graphs have no intersection points. [5]

Markscheme

METHOD 1

sketching the graph of $y = \frac{x^2}{x-3}$ ($y = x + 3 + \frac{9}{x-3}$) ***M1***

the (oblique) asymptote has a gradient equal to 1

and so the maximum value of m is 1 ***R1***

consideration of a straight line steeper than the horizontal line joining (−3, 0) and (0, 0) ***M1***

so $m > 0$ ***R1***

hence $0 < m \leq 1$ ***A1***

METHOD 2

attempting to eliminate y to form a quadratic equation in x ***M1***

$$x^2 = m\left(x^2 - 9\right)$$

$$\Rightarrow (m-1)x^2 - 9m = 0 \quad \textbf{A1}$$

EITHER

attempting to solve $-4(m-1)(-9m) < 0$ for m ***M1***

OR

attempting to solve $x^2 < 0$ *ie* $\frac{9m}{m-1} < 0 \ (m \neq 1)$ for m ***M1***

THEN

$\Rightarrow 0 < m < 1$ ***A1***

a valid reason to explain why $m = 1$ gives no solutions *eg* if $m = 1$,

$(m-1)x^2 - 9m = 0 \Rightarrow -9 = 0$ and so $0 < m \leq 1$ ***R1***

[5 marks]

3. [Maximum mark: 6]

The functions f and g are defined for $x \in \mathbb{R}$ by $f(x) = x - 2$ and $g(x) = ax + b$, where $a, b \in \mathbb{R}$.

Given that $(f \circ g)(2) = -3$ and $(g \circ f)(1) = 5$, find the value of a and the value of b. [6]

Markscheme

* This sample question was produced by experienced DP mathematics senior examiners to aid teachers in preparing for external assessment in the new MAA course. There may be minor differences in formatting compared to formal exam papers.

$(f \circ g)(x) = ax + b - 2$ **(M1)**

$(f \circ g)(2) = -3 \Rightarrow 2a + b - 2 = -3 \ (2a + b = -1)$ **A1**

$(g \circ f)(x) = a(x - 2) + b$ **(M1)**

$(g \circ f)(1) = 5 \Rightarrow -a + b = 5$ **A1**

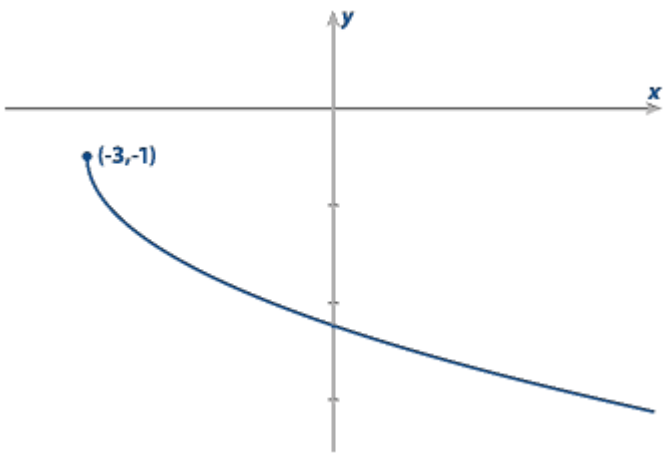
a valid attempt to solve their two linear equations for a and b **M1**

so $a = -2$ and $b = 3$ **A1**

[6 marks]

4. [Maximum mark: 14]

The following diagram shows the graph of $y = -1 - \sqrt{x+3}$ for $x \geq -3$.



(a) Describe a sequence of transformations that transforms the graph of $y = \sqrt{x}$ for $x \geq 0$ to the graph of $y = -1 - \sqrt{x+3}$ for $x \geq -3$. [3]

Markscheme

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for example,

a reflection in the x-axis (in the line $y = 0$) **A1**

a horizontal translation (shift) 3 units to the left **A1**

a vertical translation (shift) down by 1 unit **A1**

Note: Award **A1** for each correct transformation applied in a correct position in the sequence. Do not accept use of the "move" for a translation.

Note: Award **A1A1A1** for a correct alternative sequence of transformations. For example,

a vertical translation (shift) down by 1 unit, followed by a horizontal translation (shift) 3 units to the left and then a reflection in the line $y = -1$.

[3 marks]

A function f is defined by $f(x) = -1 - \sqrt{x+3}$ for $x \geq -3$.

(b) State the range of f. [1]

Markscheme

range is $f(x) \leq -1$ **A1**

Note: Correct alternative notations include $]-\infty, -1]$, $(-\infty, -1]$ or $y \leq -1$.

[1 mark]

(c) Find an expression for $f^{-1}(x)$, stating its domain. [5]

Markscheme

$-1-\sqrt{y+3}=x$ **M1**

Note: Award **M1** for interchanging x and y (can be done at a later stage).

$\sqrt{y+3}=-x-1(=-(x+1))$ **A1**

$y+3=(x+1)^2$ **A1**

so $f^{-1}(x)=(x+1)^2-3\ \left(f^{-1}(x)=x^2+2x-2\right)$ **A1**

domain is $x\leq -1$ **A1**

Note: Correct alternative notations include $]-\infty,-1]$ or $(-\infty,-1]$.

[5 marks]

(d) Find the coordinates of the point(s) where the graphs of $y=f(x)$ and $y=f^{-1}(x)$ intersect. [5]

Markscheme

the point of intersection lies on the line $y=x$

EITHER

$(x+1)^2 - 3 = x$ **M1**

attempts to solve their quadratic equation **M1**

for example, $(x+2)(x-1) = 0$ or
$x = \frac{-1 \pm \sqrt{1^2 - 4(1)(-2)}}{2}$ $\left(x = \frac{-1 \pm 3}{2}\right)$

OR

$-1 - \sqrt{x+3} = x$ **M1**

$\left(-1 - \sqrt{x+3}\right)^2 = x^2 \Rightarrow 2\sqrt{x+3} + x + 4 = x^2$

substitutes $2\sqrt{x+3} = -2(x+1)$ to obtain
$-2(x+1) + x + 4 = x^2$

attempts to solve their quadratic equation **M1**

for example, $(x+2)(x-1) = 0$ or
$x = \frac{-1 \pm \sqrt{1^2 - 4(1)(-2)}}{2}$ $\left(x = \frac{-1 \pm 3}{2}\right)$

THEN

$x = -2, 1$ **A1**

as $x \leq -1$, the only solution is $x = -2$ **R1**

so the coordinates of the point of intersection are $(-2, -2)$ **A1**

Note: Award **R0A1** if $(-2, -2)$ is stated without a valid reason given for rejecting $(1, 1)$.

[5 marks]

5. [Maximum mark: 17]

The temperature $T\,^{\circ}\mathrm{C}$ of water t minutes after being poured into a cup can be modelled by $T = T_0 \mathrm{e}^{-kt}$ where $t \geq 0$ and $T_0,\ k$ are positive constants.

The water is initially boiling at $100\,^{\circ}\mathrm{C}$. When $t = 10$, the temperature of the water is $70\,^{\circ}\mathrm{C}$.

(a) Show that $T_0 = 100$. [1]

Markscheme

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when $t = 0,\ T = 100 \Rightarrow 100 = T_0 \mathrm{e}^0$ **A1**

so $T_0 = 100$ **AG**

[1 mark]

(b) Show that $k = \frac{1}{10} \ln \frac{10}{7}$. [3]

Markscheme

correct substitution of $t = 10,\ T = 70$ **M1**

$70 = 100\mathrm{e}^{-10k}$ or $\mathrm{e}^{-10k} = \frac{7}{10}$

EITHER

$-10k = \ln \frac{7}{10}$ **A1**

$\ln \frac{7}{10} = -\ln \frac{10}{7}$ or $-\ln \frac{7}{10} = \ln \frac{10}{7}$ **A1**

OR

$\mathrm{e}^{10k} = \frac{10}{7}$ **A1**

$10k = \ln \frac{10}{7}$ **A1**

THEN

$k = \frac{1}{10} \ln \frac{10}{7}$ **AG**

[3 marks]

(c) Find the temperature of the water when $t = 15$. [2]

Markscheme

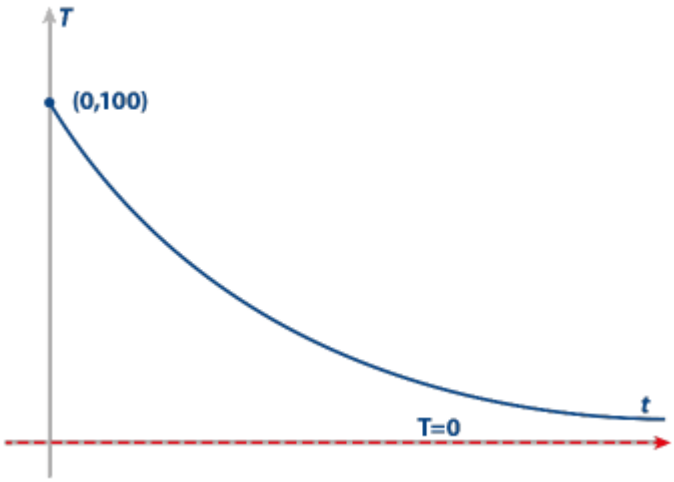
substitutes $t = 15$ into T **(M1)**

$T = 58.6\,(^\circ\mathrm{C})$ **A1**

[2 marks]

(d) Sketch the graph of T versus t, clearly indicating any asymptotes with their equations and stating the coordinates of any points of intersection with the axes. [4]

Markscheme



a decreasing exponential **A1**

starting at $(0,\ 100)$ labelled on the graph or stated **A1**

$T \to 0$ as $t \to \infty$ **A1**

horizontal asymptote $T = 0$ labelled on the graph or stated **A1**

Note: Award **A0** for stating $y = 0$ as the horizontal asymptote.

[4 marks]

(e) Find the time taken for the water to have a temperature of $50\ ^{\circ}\mathrm{C}$. Give your answer correct to the nearest second. [4]

Markscheme

$100\mathrm{e}^{-kt} = 50$ where $k = \frac{1}{10}\ln\frac{10}{7}$ **A1**

EITHER

uses an appropriate graph to attempt to solve for t **(M1)**

OR

manipulates logs to attempt to solve for t e.g. $\ln\frac{1}{2} = \left(-\frac{1}{10}\ln\frac{10}{7}\right)t$ **(M1)**

$t = \frac{\ln 2}{\frac{1}{10}\ln\frac{10}{7}} = 19.433\ldots$ **A1**

THEN

temperature will be $50\ ^\circ\mathrm{C}$ after 19 minutes and 26 seconds **A1**

[4 marks]

(f) The model for the temperature of the water can also be expressed in the form $T = T_0 a^{\frac{t}{10}}$ for $t \geq 0$ and a is a positive constant.

Find the exact value of a. [3]

Markscheme

METHOD 1

substitutes $T_0 = 100, t = 10$ and $T = 70$ into $T = T_0 a^{\frac{t}{10}}$ **(M1)**

$70 = 100a^{\frac{10}{10}}$ **A1**

$a = \frac{7}{10}$ **A1**

METHOD 2

$100a^{\frac{t}{10}} = 100\mathrm{e}^{-kt}$ where $k = \frac{1}{10}\ln\frac{10}{7}$

EITHER

$\mathrm{e}^{-k} = a^{\frac{1}{10}} \Rightarrow a = \mathrm{e}^{-10k}$ **(M1)**

OR

$a = \left(\mathrm{e}^{\left(-\frac{1}{10}\ln\frac{10}{7}\right)t}\right)^{\frac{10}{t}}$ **(M1)**

THEN

$a = \mathrm{e}^{-\ln\frac{10}{7}} \left(= \mathrm{e}^{\ln\frac{7}{10}}\right)$ **A1**

$a = \frac{7}{10}$ **A1**

[3 marks]

6. [Maximum mark: 9]

Let $f(x) = \frac{2x^2-5x-12}{x+2}$, $x \in \mathbb{R}$, $x \neq -2$.

(a) Find all the intercepts of the graph of $f(x)$ with both the x and y axes. [4]

Markscheme

$x = 0 \Rightarrow y = -6$ intercept on the y axes is (0, –6) ***A1***

$2x^2 - 5x - 12 = 0 \Rightarrow (2x+3)(x-4) = 0 \Rightarrow x = \frac{-3}{2}$ or 4 ***M1***

intercepts on the x axes are $\left(\frac{-3}{2},\ 0\right)$ and $(4,\ 0)$ ***A1A1***

[4 marks]

(b) Write down the equation of the vertical asymptote. [1]

Markscheme

$x = -2$ ***A1***

[1 mark]

(c) As $x \to \pm\infty$ the graph of $f(x)$ approaches an oblique straight line asymptote.

Divide $2x^2 - 5x - 12$ by $x + 2$ to find the equation of this asymptote. [4]

Markscheme

$f(x) = 2x - 9 + \frac{6}{x+2}$ ***M1A1***

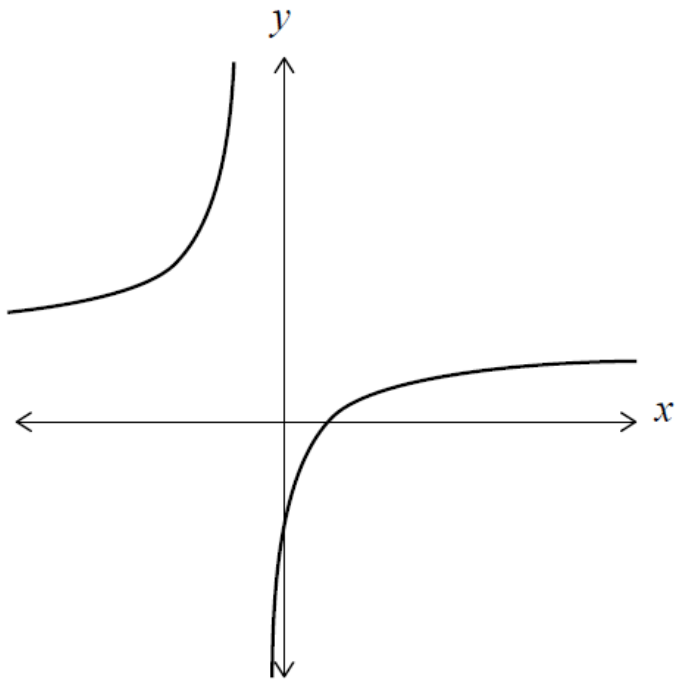
So equation of asymptote is $y = 2x - 9$ ***M1A1***

[4 marks]

7. [Maximum mark: 5]

The function f is defined by $f(x) = \frac{3x-2}{2x+1}$ for $x \in \mathbb{R}, x \neq -\frac{1}{2}$.

The following diagram shows part of the graph of $y = f(x)$.



(a) Write down the value of $f(0)$. [1]

Markscheme

-2 (accept $(0, \ -2)$) ***A1***

[1 mark]

(b) Write down the equation of the horizontal asymptote. [1]

Markscheme

$y = \frac{3}{2}$ (must be an equation) ***A1***

Note: Do not accept $\neq$ sign.

[1 mark]

The function g is defined by $g(x) = -f(x)$ for $x \geq 0$.

(c) Find the range of g. [3]

Markscheme

attempt to reflect the graph of f in the x-axis **OR** one correct value seen ***(M1)***

both correct values $-\frac{3}{2}$ and 2 (seen anywhere) ***A1***

$-\frac{3}{2} < y \leq 2$ ***A1***

[3 marks]

8. [Maximum mark: 7]

The quadratic equation $x^2 + kx + 15 - k = 0$ has two distinct real roots.

(a) Find the possible values of k. [5]

Markscheme

recognising $\Delta > 0$ (seen anywhere) ***(M1)***

$\Delta = k^2 - 4(15 - k) \ \left(= k^2 + 4k - 60\right)$ ***A1***

valid attempt to solve quadratic (in)equality ***(M1)***

$(k-6)(k+10)$ **OR** $k = \frac{-4 \pm \sqrt{4^2 - 4(-60)}}{2}$

two correct values -10 and 6 (seen anywhere) ***A1***

$k < -10,\ k > 6$ ***A1***

[5 marks]

(b) Find the possible values of k in the case where the two distinct real roots are both positive or both negative. [2]

Markscheme

recognising constant or product of roots is positive ***(M1)***

$15 - k > 0 \Rightarrow k < 15$

$k < -10,\ 6 < k < 15$ ***A1***

[2 marks]

9. [Maximum mark: 8]

The function f is defined by $f(x) = 4^x$, where $x \in \mathbb{R}$.

(a) Find $f^{-1}(8)$. Express your answer in the form $\frac{p}{q}$ where $p,\ q \in \mathbb{Z}$. [3]

Markscheme

$4^x = 8$ **OR** $f^{-1}(x) = \log_4 x$ **OR** $f^{-1}(8) = \log_4 8$ ***(A1)***

attempt to use indices with same base **OR** change of base of logs ***(M1)***

$2^{2x} = 2^3,\ 4^x = 4^{\frac{3}{2}}$ **OR** $f^{-1}(8) = \log_4 4^{\frac{3}{2}}$ **OR** $f^{-1}(8) = \frac{\log_2 8}{\log_2 4}$

$f^{-1}(8) = \frac{3}{2}$ ***A1***

[3 marks]

The function g is defined by $g(x) = 1 + \log_2 x$, where $x \in \mathbb{R}^+$.

(b.i) Find an expression for $g^{-1}(x)$. [2]

Markscheme

interchanging x and y (seen anywhere) ***(M1)***

$x = 1 + \log_2 y$ **OR** $y - 1 = \log_2 x$

$x - 1 = \log_2 y$ **OR** $2^{y-1} = x$

$g^{-1}(x) = 2^{x-1}$ (or equivalent) ***A1***

[2 marks]

(c) Show that $(f \circ g)(x) = 4x^2$. [3]

Markscheme

attempt to find composite function (in any order) ***M1***

$f(1 + \log_2 x)\ \left(= 4^{1+\log_2 x}\right)$

$4 \times 4^{\log_2 x}$ **OR** $4 \times 2^{2\log_2 x}$ **OR** $4^{\log_2 2x}$ **OR** $2^{(2+2\log_2 x)}$ **OR** $4^{(\log_4 4 + 2\log_4 x)}$ ***(A1)***

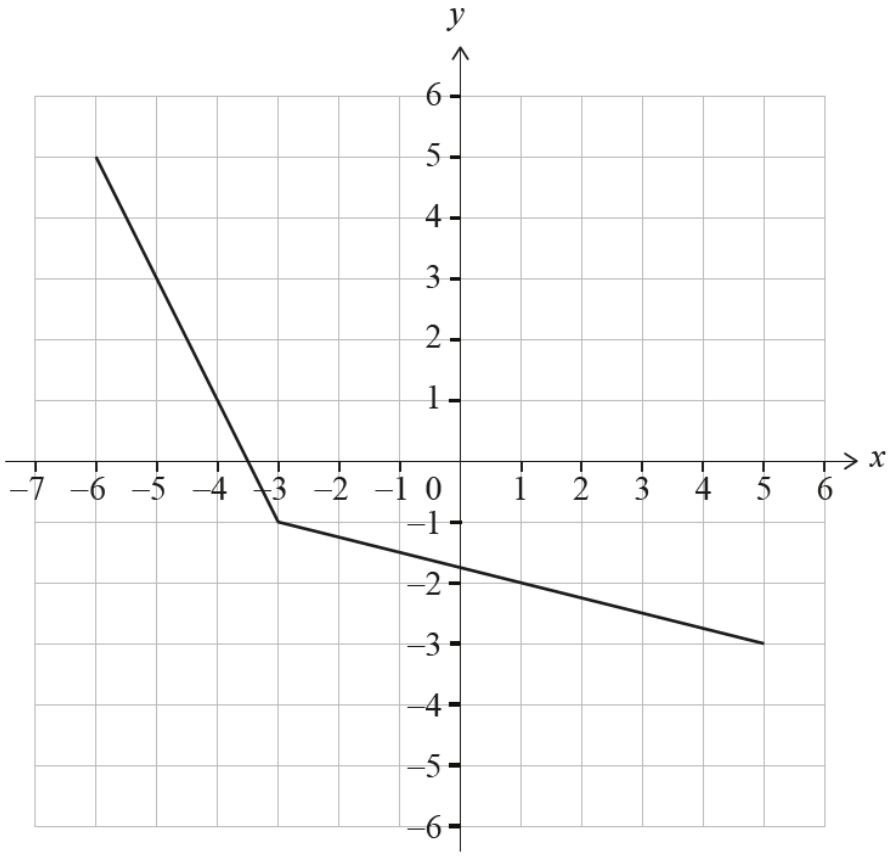
$= 4 \times 2^{\log_2 x^2}$ **OR** $2^{\log_2 (2x)^2}$ **OR** $4^{\log_4 (4x^2)}$ ***A1***

$= 4x^2$ ***AG***

[3 marks]

10. [Maximum mark: 5]

The following diagram shows the graph of $y = f(x)$, for $-6 \leq x \leq 5$.



(a) Write down the value of $f(-3)$. [1]

Markscheme

$f(-3) = -1$ *A1*

[1 mark]

(b) State the domain of f^{-1}, the inverse function of f. [1]

Markscheme

$-3 \leq x \leq 5$ *A1*

Note: Award *A1* for answers using interval notation $[-3,\ 5]$.

[1 mark]

(c) Find the value of x that satisfies $f^{-1}(2x-7) = -3$. [3]

Markscheme

$\left(f^{-1}(2x-7) = -3 \Rightarrow\right) 2x - 7 = f(-3)$ **OR** $f^{-1}(-1) = -3$ *(M1)*

$2x - 7 = -1$ *(A1)*

$x = 3$ *A1*

[3 marks]

11. [Maximum mark: 7]

Consider the curve $y = x^2 - x - 1$ and the line $y = mx - 3$, where $m \in \mathbb{R}$.

(a) Show that the curve and the line meet when $x^2 - (m+1)x + 2 = 0$. [2]

Markscheme

equating $y = mx - 3$ and $y = x^2 - x - 1$ ***M1***

$x^2 - x - 1 - (mx - 3) = 0$ **OR** $x^2 - mx - x + 2 = 0$ ***A1***

$x^2 - (m+1)x + 2 = 0$ ***AG***

[2 marks]

(b) Hence, find the values of m when the line is tangent to the curve. [5]

Markscheme

METHOD 1 (discriminant)

correct substitution into discriminant (do not award if seen only in quadratic formula) ***(A1)***

$(-(m+1))^2 - 4(1)(2)$ **OR** $(m+1)^2 - 4(1)(2)$ ***A1***

discriminant equals 0 (seen anywhere) ***A1***

$(m+1)^2 = 8$ **OR** $m^2 + 2m - 7 = 0$

$(m =) \ -1 - 2\sqrt{2}, \ -1 + 2\sqrt{2}$ ***A1A1***

METHOD 2 (derivative)

$\frac{\mathrm{d}y}{\mathrm{d}x} = 2x - 1$ ***A1***

substituting $m = 2x - 1$ or $x = \frac{m+1}{2}$ into ***AG*** from part (a) ***(A1)***

$x^2 - (2x - 1 + 1)x + 2 = 0$ **OR** $x^2 - 2x^2 + 2 = 0$ **OR** $\left(\frac{m+1}{2}\right)^2 - (m+1)\frac{m+1}{2} + 2 = 0$

$x = \pm\sqrt{2}$ **OR** $-m^2 - 2m + 7 = 0$ **OR** $m^2 + 2m - 7 = 0$ ***A1***

$(m =)\ 2\sqrt{2} - 1,\ -2\sqrt{2} - 1$ ***A1A1***

[5 marks]

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